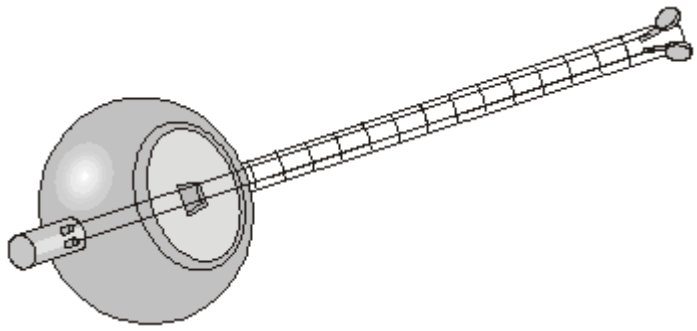
Q1.

The dotar is a musical instrument with two strings.



(a) Aftal plays the dotar very quietly.

What must he do to the strings to make a louder sound?

.....

.....

1 mark

(b) Aftal makes the strings tighter so they vibrate more quickly.

How does this affect the sound produced by the strings?
Tick the correct box.

The sound has a lower pitch.	☐
The sound is louder.	☐
The sound has a higher pitch.	☐
The sound is quieter.	☐

1 mark

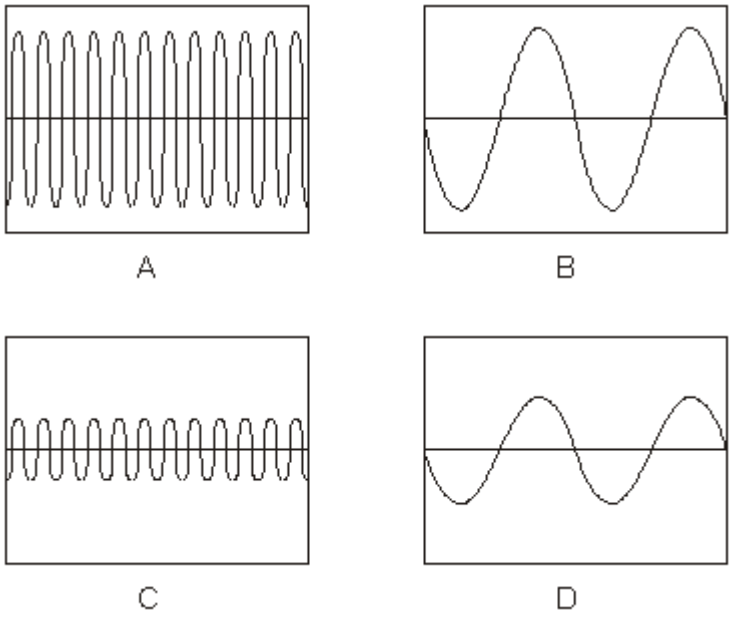
(c) One of the strings is thicker than the other, so it vibrates more slowly.

In what way is the sound made by the thicker string different from the sound made by the thinner string?

.....

1 mark

(d) Aftal played the dotar near a microphone connected to an oscilloscope.
The diagrams below show the patterns made by four sounds.



(i) How does the sound shown in trace A differ from the sound in trace B?

.....

.....

1 mark

(ii) How does the sound shown in trace A differ from the sound in trace C?

.....

.....

1 mark
maximum 5 marks

Q1.

(a) pluck them harder

*'make the strings vibrate more **or** move more*

*do **not** accept 'make the strings vibrate more quickly'*

1 (L5)

(b) The sound has a higher pitch ✓

if more than one box is ticked, award no mark

1 (L5)

(c) the sound from the thicker string has a lower pitch **or** a lower frequency

*accept 'it has a lower pitch **or** a lower frequency'*
*accept 'the sound from the thicker string is louder **or** 'it is louder'*

answers must include a comparison
'it is lower' is insufficient

1 (L5)

(d) (i) the sound in A has a higher pitch **or** a higher frequency

*accept' it has a higher pitch **or** higher frequency'*

answers must show a comparison
'the vibrations are quicker' is insufficient
'it is higher' is insufficient

1 (L6)

(ii) the sound in A is louder

accept 'it is louder'
accept 'the sound in C is quieter'
accept 'it has a greater amplitude'

'the vibrations are stronger
***or** of greater amplitude' is insufficient*
*do **not** accept 'it is higher'*

1 (L6)

[5]